

The Program in Action

Figure 58-1 shows what the output will look like when you run *rainbow.py*.

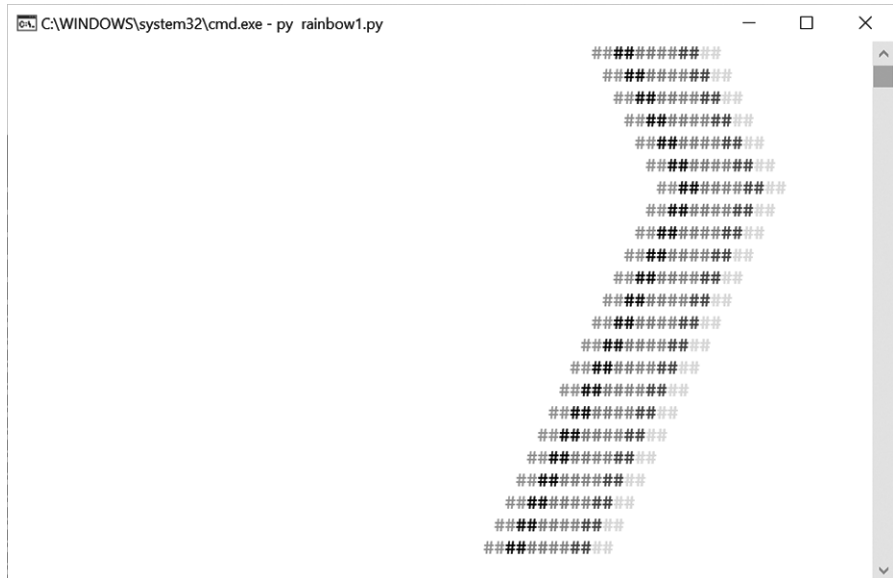


Figure 58-1: The zigzag output of the rainbow, which is in color on the screen

How It Works

This program continuously prints the same rainbow pattern. What changes is the number of space characters printed to the left of it. Increasing this number moves the rainbow to the right, and decreasing it moves the rainbow to the left. The `indent` variable keeps track of the number of spaces. The `indentIncreasing` variable is set to `True` to note that `indent` should increase until it reaches 60, at which point it changes to `False`. The rest of the code decreases the number of spaces. Once it reaches 0, it changes back to `True` again to repeat the zigzag of the rainbow.

```
1. """Rainbow, by Al Sweigart al@inventwithpython.com
2. Shows a simple rainbow animation. Press Ctrl-C to stop.
3. View this code at https://nostarch.com/big-book-small-python-projects
4. Tags: tiny, artistic, bext, beginner, scrolling"""
5.
6. import time, sys
7.
8. try:
9.     import bext
10. except ImportError:
11.     print('This program requires the bext module, which you')
12.     print('can install by following the instructions at')
13.     print('https://pypi.org/project/Bext/')
14.     sys.exit()
```